WORKED SOLUTION

STEP I 2014, Question 8

Pure • Coordinate geometry

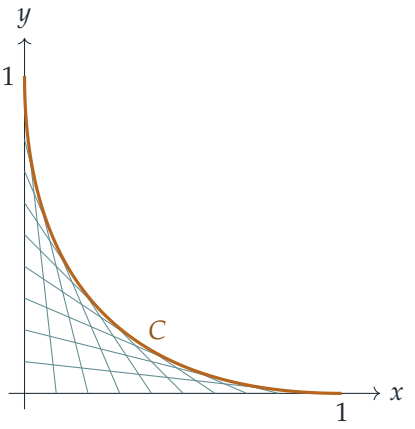
STEP I 2014, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s1-q8

The key idea

Write each line in intercept form, $\frac{x}{a} + \frac{y}{1-a} = 1$; then the intersection of two of them falls out of a pair of linear equations, and comes out symmetric in a and b. Letting $b \to a$ gives the point where a line meets its "neighbour", and these points trace out a curve which every line L_a touches. The curve C is the *envelope* of the family, and part (iii) confirms this from the other direction.



Solution

The line through $(a, 0)$ and $(0, 1-a)$ has intercepts a and $1-a$, so

$$L_a : \ \frac{x}{a} + \frac{y}{1-a} = 1, \qquad \text{that is} \qquad (1-a)x + ay = a(1-a).$$

Part (i)

We solve

$$(1-a)x + ay = a(1-a),$$
$$(1-b)x + by = b(1-b).$$

To eliminate y, multiply the first equation by b and the second by a, and subtract:

$$[b(1-a) - a(1-b)]x = ab(1-a) - ab(1-b), \qquad \text{i.e.} \qquad (b-a)x = ab(b-a).$$

Since $a \neq b$ we may divide by $b-a$, giving $x = ab$. Substituting into the first equation,

$$ay = a(1-a) - (1-a)ab = a(1-a)(1-b),$$

and since $a \neq 0$, $y = (1-a)(1-b)$. So L_a and L_b meet at

$$(ab,\ (1-a)(1-b)).$$

(The answer is symmetric in a and b, as it must be. You can check it lies on L_b too: $(1-b)ab + b(1-a)(1-b) = b(1-b)(a+1-a) = b(1-b)$.)

Part (ii)

Both coordinates are polynomials in b, so they are continuous and the limit as $b \to a$ is just the value at $b = a$:

$$(x, y) = (a^2,\ (1-a)^2).$$

Since $0 < a < 1$, we have $0 < x = a^2 < 1$, and because $a > 0$ the positive square root gives $\sqrt{x} = a$. Hence

$$y = (1-a)^2 = (1-\sqrt{x})^2,$$

so the limiting point lies on C, as required.

Part (iii)

Take any point of C. Its x-coordinate lies in $(0,1)$, so we can write it as t^2 with $t = \sqrt{x} \in (0,1)$, and the point is $(t^2, (1-t)^2)$.

Differentiating $y = (1-\sqrt{x})^2$ by the chain rule,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 2(1-\sqrt{x}) \cdot \left(-\frac{1}{2\sqrt{x}}\right) = -\frac{1-\sqrt{x}}{\sqrt{x}},$$

which at $x = t^2$ is $-\frac{1-t}{t}$. The tangent there is

$$y - (1-t)^2 = -\frac{1-t}{t}\,(x - t^2).$$

Ask yourself How do we recognise this line as one of the L_a?

A line L_a is pinned down by its two intercepts, which must be a and $1-a$. So find the intercepts of the tangent.

- Putting $y = 0$: $-(1-t)^2 = -\frac{1-t}{t}(x - t^2)$. Since $1 - t \neq 0$, we can cancel to get $x - t^2 = t(1-t)$, so $x = t$.
- Putting $x = 0$: $y = (1-t)^2 + \frac{1-t}{t}\,t^2 = (1-t)^2 + t(1-t) = 1 - t$.

So the tangent passes through $(t, 0)$ and $(0, 1-t)$ with $0 < t < 1$: it is exactly L_t. Since every point of C arises from some $t \in (0,1)$, every tangent to C is of the form L_a (with $a = \sqrt{x}$ at the point of contact). This completes the proof.

Notice that the tangent at $\left(a^2, (1-a)^2\right)$ is L_a, and this is precisely the point that part (ii) produced from L_a: each line touches C at its own limiting intersection point.

Checks

- With $a = \frac{1}{2}$, $b = \frac{1}{3}$, part (i) gives $\left(\frac{1}{6}, \frac{1}{3}\right)$. On $L_{1/2}$: $\frac{1}{2} \cdot \frac{1}{6} + \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2}$; on $L_{1/3}$: $\frac{2}{3} \cdot \frac{1}{6} + \frac{1}{3} \cdot \frac{1}{3} = \frac{2}{9} = \frac{1}{3} \cdot \frac{2}{3}$.
- At $x = \frac{1}{4}$ the curve has $y = \frac{1}{4}$ and gradient -1, so the tangent is $x + y = \frac{1}{2}$, which is $L_{1/2}$, through $(\frac{1}{2}, 0)$ and $(0, \frac{1}{2})$.
- The curve itself is part of the parabola $\sqrt{x} + \sqrt{y} = 1$, symmetric in x and y, matching the symmetry $a \leftrightarrow 1 - a$ of the family of lines.

Common slips

- Forgetting to justify $\sqrt{x} = a$ in part (ii): it needs $a > 0$, which is given.
- In part (iii), showing only that L_a touches C. The question asks the converse: start from an arbitrary tangent of C and show it is some L_a.

Beyond STEP

The curve C is the *envelope* of the family of lines L_a: the curve that touches every member. In general, the envelope of a family $F(x, y, a) = 0$ is found by eliminating a between $F = 0$ and $\partial F / \partial a = 0$, which is the calculus version of the "let $b \to a$" limit in part (ii). Envelopes appear in optics as caustics, such as the bright curve of light in a coffee cup. Siklos, *Advanced Problems in Mathematics*, has more questions in this spirit.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.